

Entanglement Goes Steady

Two independent groups have demonstrated ways to entangle quantum bits without the need for precisely timed control pulses.

By **Aziza Almanakly**

Quantum entanglement describes a link, or correlation, between the states of two or more quantum particles. For example, given a pair of entangled qubits—particles that can be in either a ground state or an excited state—measuring the state of one qubit can inform us about the state of the other. Entanglement is puzzling because it has no analogue in the classical world, where our physical intuition can be relied upon. In particular, entanglement appears to violate the principle of locality: The qubits' states remain correlated even if we move them far apart before measuring them. But entanglement is more than a curiosity: It is also critical to quantum computing, where it serves as a resource for performing quantum algorithms and remote operations between distant qubits. Now two independent research teams at the University of Illinois Urbana-Champaign (UIUC) and the Institute of Science and Technology Austria (ISTA) have each devised a new way to maintain a continuous, uninterrupted entangled link between widely separated qubits [1, 2]. The teams' demonstrations may help physicists to engineer new quantum interconnects between different processors in a

distributed architecture, eventually enabling the ultimate goal of creating a large modular quantum system that is greater than the sum of its parts [3–5].

Remote entanglement is usually generated in a quantum computer by exploiting interactions between qubits and photons. Quantum interconnects often function as channels for the propagation of these photons, which carry quantum information about the states of the interacting qubits. Most entanglement-generation procedures rely on precisely timed sequences of microwave or optical pulses, which require complex control schemes and calibration routines. The UIUC and ISTA teams proposed and demonstrated an alternative that avoids the need for such intricate timing: steady-state entanglement.

The two groups used superconducting qubits and a particular form of light–matter interaction known as driven dissipation. In both setups, entanglement is a stable state of the system, which is why these schemes are often called entanglement stabilization. Rather than choreographing the dynamics with a complex sequence of control pulses, these approaches use the interplay of a continuous microwave driving field with qubit relaxation and the subsequent photon emission into a waveguide (Fig. 1). Each experiment achieves this by leveraging a distinct phenomenon within quantum optics. Both exploit entangled, stable, so-called dark states, which are protected from waveguide dissipation via quantum interference between emission pathways. The two techniques generate entangled states between two distant qubits with modest state fidelity.

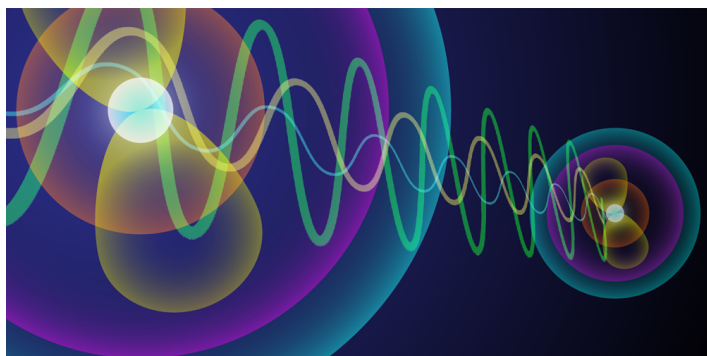


Figure 1: Two entangled qubits driven by a microwave signal. Credit: APS/Carin Cain

The UIUC team realized a unidirectional quantum system. Two qubits on separate platforms were coupled to a common waveguide, which was made unidirectional by the addition of a microwave circulator between the qubits. Thus, microwave

photons from the driving field could propagate through the waveguide in only one direction. By continuously driving both qubits, the researchers engineered a scenario where two distinct photon emission pathways coexisted: a photon emitted through the waveguide from the downstream qubit, and a photon emitted from the upstream qubit in the same direction. These two emission pathways overlapped, and by tuning the relative phases of the driving signals, they could be made to cancel each other out. The destructive quantum interference suppressed the photon emission from both qubits, forcing the qubits to remain in a stable entangled state. The upshot was driven-dissipative entanglement generation independent of distance.

The ISTA team leveraged another phenomenon of quantum optics known as squeezing. Instead of a simple classical driving field, the researchers drove the distant superconducting qubits using a squeezed field—a quantum state of light with modified uncertainty properties. In particular, they used a parametric amplifier to generate a continuous stream of entangled pairs of photons, sending individual photons from each pair to the two qubits via separate waveguides. Because these photons were inherently correlated, a photon-emission event at one qubit was quantum mechanically linked to an emission event at the other. As in the UIUC scheme, these pathways could be made to overlap and, through nonlocal destructive interference, cancel each other out, maintaining the qubits in a stable entangled state.

Although the precise principles underpinning them are different, the two schemes achieved similar outcomes. Both systems were arranged such that a stable dark state emerged, protecting the entanglement from decay into the waveguide. This dark state was populated gradually by driving the qubits continuously, resulting in steady-state remote entanglement that did not require careful timing calibration.

Whether this steady-state entanglement can remain

“always-on” during actual computational operations remains an open question. In practice, entanglement must eventually be transferred to other qubits to be consumed by an algorithm—a process that would likely be disrupted or corrupted by a continuous, active drive.

Whatever the answer to that question, these demonstrations of steady-state remote entanglement serve as a valuable proof of principle. Finding new ways to generate remote entanglement is of great interest to the quantum community for both fundamental research purposes as well as for the advancement of quantum computing and quantum sensors. In the future, a quantum-computer engineer will need to draw from a diverse toolbox of quantum interconnects with different distance scales and qubit connectivity levels depending on the specific task at hand. By providing new mechanisms for generating these quantum links, the schemes developed by the UIUC and ISTA teams—and future iterations improving these schemes—could prove to be a key resource for building distributed quantum-computing systems.

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